Chapter 11. Groups

Subgroups, homomorphisms, cyclic and linear groups

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A "new beginning": we descend to the bottom layer of abstraction, binary operations and groups, where the whole story of the book actually starts.

11.1 Binary operations and groups

Motivation. Addition of vectors, multiplication of invertible matrices and composition of rigid motions are all associative operations with an identity and inverses. Isolating exactly these properties defines a group, the first abstract algebraic structure we meet.

$X \times Y = \{(x, y) : x \in X, y \in Y\}$ is the **Cartesian product**; (x, y) is an **ordered pair**.

Definition 11.1.1 (Binary operation)

A function $* : X \times X \to X$ is a **binary operation** on X; we write $x * y$, or $x \cdot y$, xy.

Definition 11.1.2 (Associativity)

A binary operation is **associative** if $(xy)z = x(yz)$ for all $x, y, z \in X$; then xyz is meaningful.

Example 11.1.3

On $\mathfrak{M}_{n,n}(F)$, the bracket $[A, B] = AB - BA$ is not associative: for $n = 2$, $[[E_{12}, E_{21}], E_{21}] = [E_{11} - E_{22}, E_{21}] = -2E_{21}$, while $[E_{12}, [E_{21}, E_{21}]] = 0$.

From now on every binary operation is assumed associative.

Definition 11.1.4 (Group)

A set G with a binary operation is a **group** if (G1) $(xy)z = x(yz)$ for all $x, y, z \in G$; (G2) there is $e \in G$ with $xe = x = ex$ for all $x \in G$; (G3) for each $x \in G$ there is $\tilde{x} \in G$ with $x\tilde{x} = e = \tilde{x}x$.

Example 11.1.5

(a) $(\{-1, 1\}, +)$ is not a group ($1 + 1 \notin \{-1, 1\}$), but $(\{-1, 1\}, \times)$ is.
(b) $\{e\}$ with $ee = e$ is the **trivial group**.
(c) $(\mathbb{Z}, +)$ is a group; $(\mathbb{Z}, \times)$ is not.
(d) $(F, +)$ and $(F - \{0\}, \times)$ are groups; $(F, \times)$ is not.
(e) For a vector space V, $(V, +)$ is a group (axioms (V1)–(V4)).
(f) The symmetric group $(S_n, \circ)$ of Chapter 6 is a group.

Observation 11.1.6

A group is not empty.

Proof. By (G2) it contains e. ∎

Observation 11.1.7

The element e of (G2) is unique; it is the **identity** of G, written e_G.

Proof. If e' is another, $e = ee' = e'$. ■

Observation 11.1.8

For each $x \in G$, the $\tilde{x}$ of (G3) is unique; it is the **inverse** x^{-1}.

Proof. If $xx' = e = x'x$ as well, then $\tilde{x} = \tilde{x}e = \tilde{x}(xx') = (\tilde{x}x)x' = ex' = x'$. ■

Exercise 11.1.9

For $g, h \in G$: (a) $(gh)^{-1} = h^{-1}g^{-1}$;
(b) $(g^{-1})^{-1} = g$.

Solution. (a) $(gh)(h^{-1}g^{-1}) = g(hh^{-1})g^{-1} = e$ and likewise on the other side; use uniqueness of inverses.

(b) g satisfies the defining property of the inverse of g^{-1}. □

Definition 11.1.10 (Order)

G is a **finite group** if it is a finite set; then $|G|$ is its **order**. Otherwise G is an **infinite group**.

Definition 11.1.11 (Commutative group)

G is **commutative** (**abelian**) if $xy = yx$ for all $x, y \in G$.

Exercise 11.1.12

$\mu_n = \{\alpha \in \mathbb{C} : \alpha^n = 1\}$, under multiplication of complex numbers, is a finite commutative group of order n.

Solution. If $\alpha^n = \beta^n = 1$, then $(\alpha\beta)^n = 1$ and $(\alpha^{-1})^n = 1$; $1 \in \mu_n$; associativity and commutativity are inherited from $\mathbb{C}$. The roots are $e^{2\pi ik/n}$, $0 \le k < n$, which are distinct, and a polynomial of degree n has at most n roots, so $|\mu_n| = n$. □

Exercise 11.1.13

For $n \ge 3$, S_n is a finite non-commutative group of order $n!$.

Solution. $|S_n| = n!$. $(1\,2)(1\,3)$ sends $1 \mapsto 3 \mapsto 3$ but $(1\,3)(1\,2)$ sends $1 \mapsto 2$; so they differ. □

Exercise 11.1.14

On F^3 define $(a_1, a_2, a_3)^t \bullet (b_1, b_2, b_3)^t = (a_1 + b_1,\ a_2 + b_2 + a_1b_3,\ a_3 + b_3)^t$. Then $(F^3, \bullet)$ is an infinite non-commutative group.

Solution. Both $(a\bullet b)\bullet c$ and $a\bullet(b\bullet c)$ equal $(a_1+b_1+c_1,\ a_2+b_2+c_2+a_1b_3+a_1c_3+b_1c_3,\ a_3+b_3+c_3)$. The identity is 0, and the inverse of a is $(-a_1, -a_2+a_1a_3, -a_3)$. Non-commutative: $e_1\bullet e_3 = (1,1,1)$ but $e_3\bullet e_1 = (1,0,1)$. (This is the Heisenberg group of upper unitriangular 3×3 matrices in disguise.) □

11.2 Elementary properties of groups

Motivation. Powers, exponent laws, cancellation, and multiplication tables. Small groups can be classified by their tables, but whether a table is associative cannot be read off it; one needs an actual group, such as μ_n or a product, to realize it.

Notation 11.2.1

(a) (Multiplicative notation) The identity is 1; $g^1 = g$, $g^2 = gg$, …, $g^0 = 1$, $g^{-2} = g^{-1}g^{-1}$, ….
(b) (Additive notation) For an abelian group A written with $+$: identity 0, inverse $-a$, and $\ldots, (-2)a = (-a) + (-a)$, $0a = 0$, $1a = a$, $2a = a + a, \ldots$; $a - b = a + (-b)$.
(c) Commutative groups are usually written multiplicatively and abelian groups additively.
(d) Unless stated otherwise we use multiplicative notation.

Exercise 11.2.2

(a) $(g^n)^{-1} = (g^{-1})^n = g^{-n}$ for $n \in \mathbb{Z}$.
(b) Additively: $-(na) = n(-a) = (-n)a$.

Solution. $g^n(g^{-1})^n = 1$ by cancelling adjacent gg^{-1}; and $g^{-n} = (g^{-1})^n$ by definition (for $n > 0$; the other cases follow). □

Exercise 11.2.3

$(h^{-1}gh)^n = h^{-1}g^nh$ for $n \in \mathbb{Z}$.

Solution. For $n > 0$, the inner hh^{-1} cancel; $n = 0$ is clear; for $n < 0$ apply this to g^{-1}, using $(h^{-1}gh)^{-1} = h^{-1}g^{-1}h$. □

Exercise 11.2.4

Elements $ghg^{-1}h^{-1}$ are **commutators**. G is commutative iff all commutators are 1.

Solution. $ghg^{-1}h^{-1} = 1 \iff gh = hg$. □

Exercise 11.2.5

If $g^2 = 1$ for all $g \in G$, then G is commutative.

Solution. Then $g = g^{-1}$ for all g, so $ghg^{-1}h^{-1} = ghgh = (gh)^2 = 1$. □

Additively, $g^2 = 1$ reads $2a = 0$: a nonzero element can satisfy $2a = 0$ (or $3a = 0$).

Exercise 11.2.6 (Exponent laws, multiplicative)

For $g, h \in G$, $m, n \in \mathbb{Z}$: (a) $g^m g^n = g^{m+n}$;
(b) $(g^m)^n = g^{mn}$;
(c) if G is commutative, $g^n h^n = (gh)^n$.

Solution. For $m, n \geq 0$ count factors; the cases with negative exponents reduce to these using $g^{-k} = (g^{-1})^k$ and cancellation. (c) uses commutativity to regroup $(gh)(gh)\cdots$. □

Exercise 11.2.7 (Exponent laws, additive)

For $a, b \in A$, $m, n \in \mathbb{Z}$: (a) $ma + na = (m+n)a$;
(b) $m(na) = (mn)a$;
(c) $na + nb = n(a+b)$.

Solution. Exercise 11.2.6 in additive notation. □

(a) and (c) look like distributive laws; they make an abelian group a $\mathbb{Z}$-module.

Observation 11.2.8 (Cancellation law)

If $xy = xz$, then $y = z$; if $xz = yz$, then $x = y$.

Proof. $y = (x^{-1}x)y = x^{-1}(xy) = x^{-1}(xz) = z$; the other side is symmetric. ■

Exercise 11.2.9

State Observation 11.2.8 additively.

Solution. If $a + b = a + c$ then $b = c$; if $a + c = b + c$ then $a = b$. □

Example 11.2.10

$xy = x$ implies $y = 1$, since $xy = x1$.

Exercise 11.2.11

(a) If $y = u^{-1}xu$, then $x = uyu^{-1}$.
(b) If $u^{-1}xu = 1$, then $x = 1$.

Solution. (a) Multiply by u on the left and u^{-1} on the right.

(b) By (a), $x = u1u^{-1} = 1$. □

A finite group $G = \{1, x_2, x_3, \dots\}$ can be described by its **multiplication table**, whose (x_i, x_j) entry is $x_i x_j$.

Example 11.2.12

All groups of order 1 are "the same": $\{e\}$ and $(\{0\}, +)$ differ only in names.

Example 11.2.13

If $G = \{1, x\}$, then $x^2 \neq x$ (cancellation), so $x^2 = 1$: the table is forced, and G is μ_2 (or $\{\text{even}, \text{odd}\}$ under addition) up to renaming. All groups of order 2 are the same.

Observation 11.2.14

In a multiplication table every row contains each element exactly once, and so does every column.

Proof. $xy = xz$ forces $y = z$ (cancellation), so a row has no repetitions; G is finite, so each element appears. Similarly for columns. ■

Observation 11.2.15

For fixed $x \in G$, $\lambda_x : G \to G$, $\lambda_x(y) = xy$, is a bijection (it permutes G).

Proof. Injective by cancellation; surjective since $z = \lambda_x(x^{-1}z)$. ■

Example 11.2.16

If $G = \{1, x, y\}$, then $x^2 \neq x$, and $x^2 = 1$ would force $xy = y$, contradicting cancellation; so $x^2 = y$, and by Observation 11.2.14 the table is

	1	x	y
1	1	x	y
x	x	y	1
y	y	1	x

So there is at most one group of order 3 up to renaming.

Question 11.2.17

How can we check that a given table is associative?

Checking all triples is impractical, and the table alone does not tell us. But if some group is *known* to realise the table, associativity is automatic. For $\zeta = e^{2\pi i/3}$, $\mu_3 = \{1, \zeta, \zeta^2\}$, and the renaming $1 \mapsto 1$, $x \mapsto \zeta$, $y \mapsto \zeta^2$ preserves the table; so the table of Example 11.2.16 is associative because multiplication of complex numbers is. (This shows how much Exercise 11.1.12, which rests on the existence of $\mathbb{C}$, really says.)

Example 11.2.18

For $G = \{1, x, y, z\}$, cancellation and Observation 11.2.14 leave four possible tables (up to symmetry): one in which every element squares to 1, and three in which one non-identity element squares to 1 (the second, third and fourth tables), which are $\mu_4 = \{1, -1, i, -i\}$ up to renaming.

Exercise 11.2.19

The second, third and fourth tables of Example 11.2.18 are associative.

Solution. Each is the table of μ_4 after renaming its elements (for instance $x \mapsto -1$, $y \mapsto i$, $z \mapsto -i$, with appropriate choices); multiplication in μ_4 is associative. □

Example 11.2.20

For groups G_1, G_2, the Cartesian product $G_1 \times G_2$ with $(g_1, g_2)(h_1, h_2) = (g_1h_1, g_2h_2)$ is a group, the **direct product**.

Example 11.2.21

$\mu_2 \times \mu_2$ realises the first table of Example 11.2.18; e.g. $(1, -1)(1, -1) = (1, 1)$. It is the **Klein 4-group**. It is not the same as μ_4: in $\mu_2 \times \mu_2$ every element squares to the identity, in μ_4 not.

Exercise 11.2.22

(If you can,) show with tables that a group of order 5 is μ_5 up to renaming.

Solution [Supplement] (via Lagrange's theorem, Observation 12.1.10). Let $x \neq 1$. The powers of x form a subgroup $\langle x \rangle$ (§11.3) whose order divides 5, hence equals 5. So $G = \{1, x, x^2, x^3, x^4\}$ with $x^5 = 1$, and $x^k \mapsto e^{2\pi ik/5}$ is a renaming onto μ_5 preserving the table. □

Exercise 11.2.23

If $|G| \leq 5$, then G is commutative; so S_3 is the smallest non-commutative group.

Solution. By Examples 11.2.12, 11.2.13, 11.2.16, 11.2.18 and Exercise 11.2.22, every such group is μ_n or $\mu_2 \times \mu_2$, whose tables are symmetric. S_3 has order 6 and is non-commutative. □

Remark 11.2.24

Semigroups, monoids, and one-sided identities and inverses appear in other texts; they are not needed here, nor is the classification of finite groups.

11.3 Subgroups

Motivation. Subgroups play the role of subspaces. Closure under products and inverses suffices. Intersections of subgroups are subgroups, which gives the subgroup generated by a set; it consists of all words in the generators and their inverses.

Definition 11.3.1 (Subgroup)

A subset $H \subseteq G$ that is a group under the operation of G is a **subgroup**, $H \leq G$.

Observation 11.3.2

Let $H \leq G$. (a) $e_H = e_G$.
(b) The inverse of $h \in H$ in H equals its inverse h^{-1} in G.

Proof. (a) In G, $e_H e_H = e_H = e_H e_G$; cancel e_H.

(b) In G, $hh' = e_H = e_G = hh^{-1}$; cancel h. ■

Exercise 11.3.3

Why was $0_W = 0_V$ for subspaces $W \leq V$ never an issue?

Solution. Because $0_W = 0 \cdot w$ for any $w \in W$, computed in V, which is 0_V: scalar multiplication pins down the zero vector. □

Observation 11.3.4

$H \subseteq G$ is a subgroup iff (S1) $h_1, h_2 \in H \Rightarrow h_1 h_2 \in H$; (S2) $e \in H$; (S3) $h \in H \Rightarrow h^{-1} \in H$.

Proof. Necessity: the operation must restrict to H (S1), and Observation 11.3.2 gives (S2), (S3). Sufficiency: associativity is inherited from G. ■

Exercise 11.3.5

A nonempty $H \subseteq G$ is a subgroup iff $h_1, h_2 \in H \Rightarrow h_1 h_2^{-1} \in H$.

Solution. Necessity is clear. For sufficiency, take $h \in H$: then $e = hh^{-1} \in H$, $h^{-1} = eh^{-1} \in H$, and $h_1 h_2 = h_1 (h_2^{-1})^{-1} \in H$. □

Example 11.3.6

(a) $\{e\} \leq G \leq G$. (b) Under addition, $\mathbb{Z} < \mathbb{Q} < \mathbb{R} < \mathbb{C}$. (c) Under multiplication, $\mu_2 < \mathbb{Q} - \{0\} < \mathbb{R} - \{0\} < \mathbb{C} - \{0\}$. (d) The alternating group A_n is a subgroup of S_n. (e) If $m \mid n$, then $\mu_m \leq \mu_n$. (f) If W is a subspace of V, then $(W, +) \leq (V, +)$.

Observation 11.3.7

If $K \leq H$ and $H \leq G$, then $K \leq G$.

Proof. K is a group under the operation of H, which is that of G. ■

Exercise 11.3.8

If $I \neq \emptyset$ and $H_i \leq G$ for all $i \in I$, then $\bigcap_i H_i \leq G$.

Solution. Each H_i contains e and is closed under products and inverses; so is the intersection. □

Notation 11.3.9

For $H, K \subseteq G$, $HK = \{hk : h \in H, k \in K\}$.

Exercise 11.3.10

(a) If G is commutative and $H, K \leq G$, then $HK \leq G$.
(b) In $S_3 = \{id, (1\,2), (1\,3), (2\,3), (1\,2\,3), (1\,3\,2)\}$, $H = \{id, (1\,2)\}$ and $K = \{id, (2\,3)\}$ are subgroups but $HK \not\leq S_3$.

Solution. (a) $(h_1 k_1)(h_2 k_2)^{-1} = h_1 h_2^{-1} k_1 k_2^{-1} \in HK$ by commutativity; apply Exercise 11.3.5. (b) $HK = \{id, (2\,3), (1\,2), (1\,2)(2\,3)\}$ has four elements, while KH contains $(2\,3)(1\,2) \neq (1\,2)(2\,3)$. If HK were a subgroup it would contain $(2\,3)(1\,2) = ((1\,2)(2\,3))^{-1}$, which is not in HK. □

Definition 11.3.11 (Generated subgroup)

For $S \subseteq G$, the smallest subgroup of G containing S is the **subgroup generated by** S, $\langle S \rangle$: $S \subseteq \langle S \rangle \leq G$, and $S \subseteq H \leq G$ implies $\langle S \rangle \subseteq H$.

Observation 11.3.12

$\langle S \rangle = \bigcap_{S \subseteq H \leq G} H$; in particular $\langle S \rangle$ exists and is unique.

Proof. By Exercise 11.3.8 the intersection is a subgroup containing S, and it lies in every subgroup containing S. ■

Example 11.3.13

$\langle\emptyset\rangle = \{e\}$, and for $x \in G$, $\langle x\rangle = \{x^i : i \in \mathbb{Z}\}$ (this set is a subgroup by the exponent laws, and any subgroup containing x contains it). It need not be infinite: $x^i = x^j$ is possible for $i \neq j$.

Definition 11.3.14 (Cyclic group)

(a) $\langle x\rangle$ is the **cyclic subgroup** generated by x.
(b) If $G = \langle x\rangle$ for some x, G is a **cyclic group** with generator x. Cyclic groups are commutative (§11.7).

Definition 11.3.15 (Alphabet and words)

(a) For $S \subseteq G$ let $S^{-1} = \{s^{-1} : s \in S\}$; $S \cup S^{-1}$ is the **alphabet**.
(b) Finite products of letters are **words**; e is the empty word. (Different words may represent the same element.)

Observation 11.3.16

The set of all words in the alphabet $S \cup S^{-1}$ is a subgroup of G.

Proof. It contains the empty word, products of words are words, and $(s_1 \cdots s_k)^{-1} = s_k^{-1} \cdots s_1^{-1}$ is a word (Exercise 11.1.9); apply Observation 11.3.4. ■

Proposition 11.3.17

$\langle S\rangle$ is the set K of all words in the alphabet $S \cup S^{-1}$.

Proof. $\langle S\rangle$ is a subgroup containing S, so it contains all such words: $K \subseteq \langle S\rangle$. Conversely K is a subgroup (Observation 11.3.16) containing S, so $\langle S\rangle \subseteq K$. ■

Exercise 11.3.18

(a) $(\mathbb{Z}, +)$ is cyclic; how many generators does it have?
(b) $3\mathbb{Z} = \{3a : a \in \mathbb{Z}\}$ is a cyclic subgroup of $(\mathbb{Z}, +)$.
(c) $\langle 2, 3\rangle = \mathbb{Z} = \langle 4, 15\rangle$, but $\langle 6, 9\rangle \neq \mathbb{Z}$.

Solution. (a) $\mathbb{Z} = \langle 1\rangle = \langle -1\rangle$, and these are the only generators: $\langle n\rangle = n\mathbb{Z} \neq \mathbb{Z}$ for $|n| \neq 1$.

(b) $3\mathbb{Z} = \langle 3\rangle$.

(c) $1 = 3 - 2 = 4 \cdot 4 - 15$, so 1 lies in both; but every element of $\langle 6, 9\rangle$ is a multiple of 3, so $\langle 6, 9\rangle = 3\mathbb{Z}$. □

Exercise 11.3.19

(a) μ_5 is cyclic; how many generators?
(b) μ_6 is cyclic; how many generators?

Solution. $\mu_n = \langle\zeta\rangle$ with $\zeta = e^{2\pi i/n}$, and ζ^k generates iff $\gcd(k, n) = 1$. (a) 4 generators ($k = 1, 2, 3, 4$).

(b) 2 generators ($k = 1, 5$). □

Exercise 11.3.20

(a) Find all subgroups of S_3.
(b) Is S_3 cyclic?
(c) $\langle (1\,2), (2\,3)\rangle = S_3 = \langle (1\,2), (1\,2\,3)\rangle$.

Solution. (a) $\{id\}$, $\langle (1\,2)\rangle$, $\langle (1\,3)\rangle$, $\langle (2\,3)\rangle$, $A_3 = \langle (1\,2\,3)\rangle$, S_3. (Any subgroup containing two distinct transpositions, or a transposition and a 3-cycle, contains their products and is all of S_3, by (c).)

(b) No: the elements have orders $1, 2, 3$, so no single element generates the 6 elements.

(c) $\langle (1\,2), (2\,3)\rangle$ contains $(1\,2)(2\,3)$ and $(2\,3)(1\,2)$, the two 3-cycles, and $(1\,2)(2\,3)(1\,2) = (1\,3)$: all six elements. Likewise $(1\,2)$ and $(1\,2\,3)$ generate $(1\,2\,3)^2$ and the products $(1\,2)(1\,2\,3)$, $(1\,2\,3)(1\,2)$, which are the other transpositions. □

11.4 The heart of undergraduate algebra

Motivation. Every subgroup of $(\mathbb{Z}, +)$ is $n\mathbb{Z}$, by the division algorithm applied to its least positive element. The same argument, with polynomials in place of integers, gave the uniqueness of the minimal polynomial; it reappears in many guises.

Example 11.4.1

For $n \in \mathbb{Z}$, $n\mathbb{Z} = \{na : a \in \mathbb{Z}\} = \langle n\rangle$ is a subgroup of $(\mathbb{Z}, +)$.

Theorem 11.4.2 (The heart of undergraduate algebra)

The subgroups of $(\mathbb{Z}, +)$ are exactly the $n\mathbb{Z}$, $0 \le n \in \mathbb{Z}$.

Proof. Let $H \le \mathbb{Z}$. If $H = \{0\}$, then $H = 0\mathbb{Z}$. Otherwise H contains a positive integer (with $-a$ for a), and we let n be the least positive element of H (well-ordering). For $a \in H$, divide: $a = qn + r$ with $0 \le r < n$. Then $qn \in H$, so $r = a - qn \in H$, and minimality forces $r = 0$. Hence $H \le n\mathbb{Z}$, and $n\mathbb{Z} = \langle n\rangle \le H$. ■

11.5 Group isomorphisms

Motivation. Groups that differ only in the names of their elements, with "rename then multiply = multiply then rename", are isomorphic. Everything a group theorist studies is invariant under isomorphism.

Definition 11.5.1 (Isomorphism)

Groups G, G' are **isomorphic**, $G \approx G'$, if there is a bijection $\varphi : G \to G'$ with $\varphi(g_1 g_2) = \varphi(g_1)\varphi(g_2)$; φ is a (**group**) **isomorphism**. Equivalently, with $(\varphi \times \varphi)(g_1, g_2) = (\varphi g_1, \varphi g_2)$, the square

$$\begin{array}{ccc} G \times G & \xrightarrow{*} & G \\ {\scriptstyle \varphi\times\varphi}\downarrow & & \downarrow{\scriptstyle \varphi} \\ G' \times G' & \xrightarrow[*']{} & G' \end{array}$$

commutes.

Example 11.5.2

Let $\varphi : G \to H$ be an isomorphism, $g \in G$, $S \subseteq G$. Then (a) $\varphi(e_G) = e_H$;
(b) $\varphi(g^{-1}) = \varphi(g)^{-1}$;
(c) if $\langle S \rangle = G$, then $\langle \varphi(S) \rangle = H$;
(d) if $g^2 = e_G$, then $\varphi(g)^2 = e_H$;
(e) if G is commutative, so is H;
(f) if G is cyclic, so is H;
(g) if G has six subgroups, so does H.

Proof [Supplement]. (a) $\varphi(e)\varphi(e) = \varphi(e) = \varphi(e)e_H$; cancel.

(b) $\varphi(g)\varphi(g^{-1}) = \varphi(e) = e_H$.

(c) φ maps words in $S \cup S^{-1}$ onto words in $\varphi S \cup (\varphi S)^{-1}$ (Proposition 11.3.17) and is onto.

(d) Apply φ.

(e) $\varphi(a)\varphi(b) = \varphi(ab) = \varphi(ba) = \varphi(b)\varphi(a)$, and φ is onto.

(f) Case of (c).

(g) $K \mapsto \varphi(K)$ is a bijection between the sets of subgroups (with inverse $K' \mapsto \varphi^{-1}(K')$). ■

Group theory is the study of properties invariant under isomorphism.

Exercise 11.5.3

Is $\mathbb{Z} \approx \mathbb{Z} \times \mathbb{Z}$ (additive groups)?

Solution. No. $\mathbb{Z}$ is cyclic, so its isomorphic image would be cyclic (Example 11.5.2(f)). But $\mathbb{Z} \times \mathbb{Z}$ is not cyclic: $\langle (a, b) \rangle = \{(na, nb)\}$ never contains both $(1, 0)$ and $(0, 1)$. □

Example 11.5.4

Properties *not* preserved by isomorphisms: "the operation is called multiplication", "G has an element named x", "the elements are complex numbers", "$|g| = 7$ as a real number", "g integrates to h".

Observation 11.5.5

$\approx$ is an equivalence relation.

Proof. Exercise 11.5.6. ■

Exercise 11.5.6

(a) id_G is an isomorphism.
(b) If $\varphi : G \to H$ is an isomorphism, so is φ^{-1}.
(c) If $\psi : K \to G$, $\varphi : G \to H$ are isomorphisms, so is $\varphi \circ \psi$.

Solution. (b) With $h_i = \varphi(g_i)$: $\varphi^{-1}(h_1 h_2) = \varphi^{-1}(\varphi(g_1 g_2)) = g_1 g_2 = \varphi^{-1}(h_1)\varphi^{-1}(h_2)$. (a), (c) are direct. □

Example 11.5.7

$\exp : (\mathbb{R}, +) \to (\mathbb{R}^+, \times)$ is a group isomorphism, with inverse log: $e^{a+b} = e^a e^b$.

Remark 11.5.8

For vector spaces, isomorphism is decided by the dimension. For groups, deciding whether two given groups are isomorphic (the *isomorphism problem*) is in general hard.

11.6 Group homomorphisms

Motivation. Homomorphisms are the structure-preserving maps of groups, the analogue of linear maps. Kernels, images and injectivity work as for linear maps.

Definition 11.6.1 (Homomorphism)

$\varphi : G \to H$ is a (**group**) **homomorphism** if $\varphi(xy) = \varphi(x)\varphi(y)$ for all $x, y \in G$.

Observation 11.6.2

For a homomorphism φ: (a) $\varphi(e) = e$;
(b) $\varphi(x^{-1}) = \varphi(x)^{-1}$.

Proof. As in Example 11.5.2(a), (b). ■

Definition 11.6.3 (Kinds of homomorphisms)

A homomorphism $\varphi : G \to H$ is a **monomorphism** if injective, an **epimorphism** if surjective, an **isomorphism** if bijective, an **endomorphism** if $G = H$, and an **automorphism** if it is a bijective endomorphism.

Observation 11.6.4

A homomorphism $\varphi : G \to H$ is an isomorphism iff there is a homomorphism $\psi : H \to G$ with $\psi\varphi = id_G$ and $\varphi\psi = id_H$.

Proof. If φ is an isomorphism, take $\psi = \varphi^{-1}$ (Exercise 11.5.6). Conversely such a ψ is an inverse function, so φ is bijective. ■

Definition 11.6.5 (Kernel and image)

$\ker \varphi = \varphi^{-1}(e) = \{x \in G : \varphi(x) = e\}$ and $\operatorname{im} \varphi = \varphi(G)$.

Observation 11.6.6

For a homomorphism $\varphi : G \to H$: (a) $\ker \varphi \leq G$ and $\operatorname{im} \varphi \leq H$;
(b) if φ is a monomorphism, $G \approx \operatorname{im} \varphi$;
(c) if $K \leq G$, then $\varphi(K) \leq H$.

Proof [Supplement]. (a) If $\varphi x = \varphi y = e$, then $\varphi(xy^{-1}) = e$, and $e \in \ker \varphi$ (Exercise 11.3.5).
(c) $\varphi(x)\varphi(y)^{-1} = \varphi(xy^{-1}) \in \varphi(K)$; the case $K = G$ gives $\operatorname{im} \varphi \leq H$.
(b) $\varphi : G \to \operatorname{im} \varphi$ is a bijective homomorphism. ■

Observation 11.6.7

For a homomorphism φ the following are equivalent: (1) φ is a monomorphism; (2) $\varphi x = \varphi y \Rightarrow x = y$; (3) $\varphi x = \varphi e = e \Rightarrow x = e$; (4) $\ker \varphi = \{e\}$.

Proof. All but (3)⇒(2) are immediate. If (3) holds and $\varphi x = \varphi y$, then $\varphi(xy^{-1}) = \varphi(x)\varphi(y)^{-1} = e$, so $xy^{-1} = e$, $x = y$. ■

Observation 11.6.8

For a homomorphism $\varphi : G \to H$ and $S \subseteq G$: (a) $\varphi\langle S\rangle = \langle \varphi S\rangle$;
(b) if $G = \langle S\rangle$, then $\operatorname{im}\varphi = \langle \varphi S\rangle$;
(c) φ is an epimorphism iff it maps a generating set of G to a generating set of H.

Proof. (a) Elements of $\langle S\rangle$ are words in $S \cup S^{-1}$ (Proposition 11.3.17); their images are words in $\varphi S \cup (\varphi S)^{-1}$, so $\varphi\langle S\rangle \leq \langle \varphi S\rangle$. Conversely $\varphi\langle S\rangle$ is a subgroup of H (Observation 11.6.6(c)) containing φS, so $\langle \varphi S\rangle \leq \varphi\langle S\rangle$.

(b) The case $\langle S\rangle = G$.

(c) By (b), $\operatorname{im}\varphi = H$ iff $\langle \varphi S\rangle = H$. ■

Exercise 11.6.9

If $\varphi : G \to H$ is a homomorphism and G is cyclic, then $\operatorname{im}\varphi$ is cyclic.

Solution. If $G = \langle x\rangle$, then $\operatorname{im}\varphi = \langle \varphi x\rangle$ by Observation 11.6.8(b). □

Exercise 11.6.10

Let $\varphi(g) = g^{-1}$ and $\psi(g) = g^2$ on G. (a) φ is a bijection; is it a homomorphism?
(b) Is ψ a homomorphism?

Solution. (a) φ is its own inverse. It is a homomorphism iff $(xy)^{-1} = x^{-1}y^{-1}$, i.e. $y^{-1}x^{-1} = x^{-1}y^{-1}$ for all x, y, iff G is commutative.

(b) $\psi(xy) = \psi(x)\psi(y)$ iff $xyxy = xxyy$, iff $yx = xy$; so again iff G is commutative. □

Exercise 11.6.11

(a) For $n \geq 2$, $\operatorname{sgn} : S_n \to \mu_2$ is an epimorphism.
(b) Describe $\ker(\operatorname{sgn})$.

Solution. (a) $\operatorname{sgn}(\sigma\tau) = \operatorname{sgn}\sigma \operatorname{sgn}\tau$, and $\operatorname{sgn}(1\,2) = -1$.

(b) $\ker(\operatorname{sgn}) = A_n$, the even permutations. □

11.7 Cyclic groups

Motivation. The simplest groups: up to isomorphism a cyclic group is $\mathbb{Z}$ or μ_n. The order of an element is read off from the kernel of $a \mapsto x^a$, another avatar of "the heart of undergraduate algebra".

Example 11.7.1

(a) $\mathbb{Z} = \langle 1 \rangle$ is an infinite cyclic group, and so is $n\mathbb{Z} = \langle n \rangle$ for $n > 0$; $\varphi(a) = na$ is an isomorphism $\mathbb{Z} \approx n\mathbb{Z}$, so unlike a finite-dimensional vector space $\mathbb{Z}$ is isomorphic to a proper subgroup.
(b) For $\zeta = e^{2\pi i/n}$, $\mu_n = \{1, \zeta, \ldots, \zeta^{n-1}\} = \langle \zeta \rangle$ is a finite cyclic group of order n (the n-th roots of unity on the unit circle).

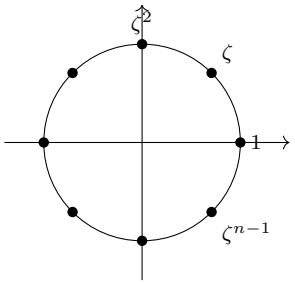


Figure. *μ_n ($n = 8$): the vertices of a regular n-gon on the unit circle.*

Definition 11.7.2 (Order of an element)

The **order** of $x \in G$ is $|x| = |\langle x \rangle|$.

Observation 11.7.3

If x has finite order, then $|x| = n$ iff (1) $x^n = 1$ and (2) $x^m \neq 1$ for $0 < m < n$; i.e. n is the least positive integer with $x^n = 1$.

Proof. $\langle x \rangle$ is finite, so $x^i = x^j$ for some $i < j$, and $x^{j-i} = 1$: the set $\{a \geq 1 : x^a = 1\}$ is nonempty. Let n be its least element. Then $\langle x \rangle = \{1, x, \ldots, x^{n-1}\}$: for $b \in \mathbb{Z}$, $b = qn + r$ with $0 \leq r < n$ and $x^b = (x^n)^q x^r = x^r$. These n powers are distinct, since $x^k = x^\ell$ with $0 \leq k < \ell < n$ would give $x^{\ell-k} = 1$ with $0 < \ell - k < n$. So $|x| = n$. Conversely, if n satisfies (1) and (2), it is the least such integer, and the same argument gives $|\langle x \rangle| = n$. ■

Definition 11.7.4 (Order, second definition)

For $x \in G$, $\gamma_x : \mathbb{Z} \to G$, $\gamma_x(a) = x^a$, is a homomorphism (exponent laws), so $\ker \gamma_x = n\mathbb{Z}$ for a unique $n \geq 0$ (Theorem 11.4.2). Set $|x| = \infty$ if $n = 0$ and $|x| = n$ otherwise.

Exercise 11.7.5

Definitions 11.7.2 and 11.7.4 agree.

Solution. If $\ker \gamma_x = 0$, γ_x is injective and $\langle x \rangle = \operatorname{im} \gamma_x$ is infinite. If $\ker \gamma_x = n\mathbb{Z}$, $n > 0$, then n is the least positive integer with $x^n = 1$, and $|\langle x \rangle| = n$ by Observation 11.7.3. □

Exercise 11.7.6

(a) $|x| = 1$ iff $x = 1$.
(b) $|x| = 2$ iff $1 \neq x = x^{-1}$.
(c) Every nonzero element of $(\mathbb{R}, +)$ has order ∞.
(d) A k-cycle in S_n has order k.

Solution. (a),

(b) by Observation 11.7.3.

(c) $na \neq 0$ for $n \neq 0$, $a \neq 0$.

(d) A k-cycle $(i_1 \ldots i_k)$ to the power m sends $i_1 \mapsto i_{1+m}$ (indices mod k), which is i_1 only if $k \mid m$, and the k-th power is the identity. □

Exercise 11.7.7

If $|x| = n$ and $a, b \in \mathbb{Z}$, then $x^a = x^b$ iff $n \mid (a - b)$.

Solution. $x^a = x^b \iff x^{a-b} = 1 \iff a - b \in \ker \gamma_x = n\mathbb{Z}$. □

Exercise 11.7.8

Every subgroup of a cyclic group is cyclic.

Solution. Let $G = \langle x \rangle$, $H \leq G$. Then $\gamma_x : \mathbb{Z} \to G$ is onto, $\gamma_x^{-1}(H) \leq \mathbb{Z}$ equals $m\mathbb{Z}$ (Theorem 11.4.2), and $H = \gamma_x(m\mathbb{Z}) = \langle x^m \rangle$. □

Exercise 11.7.9

Up to isomorphism the cyclic groups are $\mathbb{Z}$ and the μ_n.

Solution. If $G = \langle x \rangle$ and $\ker \gamma_x = 0$, then $\gamma_x : \mathbb{Z} \to G$ is an isomorphism. If $\ker \gamma_x = n\mathbb{Z}$, $n > 0$, define $\varphi : G \to \mu_n$, $\varphi(x^a) = \zeta^a$. It is well defined and injective because $x^a = x^b \iff n \mid a - b \iff \zeta^a = \zeta^b$ (Exercise 11.7.7), it is onto, and it is a homomorphism by the exponent laws. □

11.8 Examples of groups and homomorphisms

Motivation. A catalogue: symmetric groups and their presentation by generators and relations, Cayley's theorem (every group is a permutation group), conjugacy, automorphisms and products.

Exercise 11.8.1

(a) Composites of homomorphisms are homomorphisms.
(b) If $L \in \mathfrak{L}(V, W)$, then $L : (V, +) \to (W, +)$ is a homomorphism.

Solution. (a) $\varphi\psi(xy) = \varphi(\psi x \psi y) = \varphi\psi x \, \varphi\psi y$.

(b) $L(v + w) = Lv + Lw$. □

Exercise 11.8.2

(a) The constant map $\mathbf{1}(g) = 1$ is a homomorphism, the **trivial map** (written 0 for abelian groups). (b) For $H \leq G$, the inclusion $\iota_H : H \to G$ is a monomorphism, the **natural embedding**. (c) For $H \leq G$ and a homomorphism $\varphi : G \to G'$, the restriction $\varphi|_H$ is a homomorphism.

Solution. All follow directly from the definitions ($1 \cdot 1 = 1$; the operation of H is that of G). □

Example 11.8.3

"Every permutation is a product of transpositions" says S_n is *generated* by transpositions. $\operatorname{sgn} : S_n \to \mu_2$ is a homomorphism with $\ker(\operatorname{sgn}) = A_n$, so $A_n \leq S_n$ automatically (Observation 11.6.6).

Example 11.8.4

(a) In S_3, put $\sigma = (1\,2)$, $\tau = (1\,2\,3)$. Then $\sigma^2 = id = \tau^3$, $\tau^2 = (1\,3\,2)$, $\sigma\tau = (2\,3)$, $\sigma\tau^2 = (1\,3)$, so

$$S_3 = \{id, \tau, \tau^2, \sigma, \sigma\tau, \sigma\tau^2\} = \{\sigma^i\tau^j : 0 \leq i \leq 1,\ 0 \leq j \leq 2\} = \langle \sigma, \tau \rangle.$$

(b) Moreover $\sigma^{-1}\tau\sigma = \tau^2 = \tau^{-1}$, i.e. $\tau\sigma = \sigma\tau^2$. This rule moves σ to the left, so the three relations

$$\sigma^2 = 1, \qquad \tau^3 = 1, \qquad \tau\sigma = \sigma\tau^2$$

determine the whole multiplication table of S_3; e.g. $(\sigma\tau)(\sigma\tau^2) = \sigma(\tau\sigma)\tau^2 = \sigma\sigma\tau^2\tau^2 = \tau$.
(c) This is the plane rule "flip, rotate, flip back = rotate backwards" (Observation 9.5.9); see Example 11.9.23.

Definition 11.8.5 (Symmetric group on a set)

For a nonempty set X, $S_X = \{\sigma : X \to X \text{ bijection}\}$ is a group under composition, the **symmetric group** on X.

Exercise 11.8.6

(a) How should "X and Y are the same as sets" be defined?
(b) If X, Y are isomorphic as sets, then $S_X \approx S_Y$.

Solution. (a) There is a bijection $f : X \to Y$ (a renaming).

(b) $\sigma \mapsto f \circ \sigma \circ f^{-1}$ is a bijection $S_X \to S_Y$ with inverse $\tau \mapsto f^{-1}\tau f$, and $f\sigma\sigma' f^{-1} = (f\sigma f^{-1})(f\sigma' f^{-1})$. □

Exercise 11.8.7

(a) The **automorphism group** $\operatorname{Aut}(G) = \{\varphi : G \to G \text{ automorphism}\}$ is a subgroup of S_G.
(b) For $g \in G$, $\operatorname{Int}(g)(x) = gxg^{-1}$ defines $\operatorname{Int}(g) \in \operatorname{Aut}(G)$, an **inner automorphism**; $\operatorname{Int}(G) = \{\operatorname{Int}(g) : g \in G\} \leq \operatorname{Aut}(G)$.
(c) (Remark) Describing $\operatorname{Aut}(G)$ in general is hard; one first asks whether there are **outer** automorphisms (not inner).

Solution. (a) Composites and inverses of automorphisms are automorphisms (Exercise 11.5.6), and $id \in \operatorname{Aut}(G)$. (b) $\operatorname{Int}(g)(xy) = gxyg^{-1} = gxg^{-1}gyg^{-1}$. It is bijective with inverse $\operatorname{Int}(g^{-1})$. And $\operatorname{Int}(g) \circ \operatorname{Int}(h) = \operatorname{Int}(gh)$, so $\operatorname{Int}(G)$ is closed under composition and inverses. □

Theorem 11.8.8 (Cayley)

$\lambda : G \to S_G$, $\lambda(x) = \lambda_x$ with $\lambda_x(g) = xg$, is a monomorphism. Hence every group is isomorphic to a subgroup of a symmetric group.

Proof. $\lambda_x \in S_G$ by Observation 11.2.15. For $x, y, g \in G$, $\lambda_{xy}(g) = xyg = \lambda_x(\lambda_y(g))$, so $\lambda_{xy} = \lambda_x \circ \lambda_y$ and λ is a homomorphism. If $\lambda_x = id$, then $x = \lambda_x(1) = 1$, so $\ker\lambda = \{1\}$ (Observation 11.6.7). By

Observation 11.6.6(b), $G \approx \operatorname{im}\lambda \leq S_G$. ■

(With right multiplication $\rho_x(g) = gx$ one would get $\rho_{xy} = \rho_y \circ \rho_x$, an anti-homomorphism.) S_G is far bigger than G, so Cayley's theorem is of limited practical use.

Definition 11.8.9 (Conjugacy)

$x \sim y$ iff $y = g^{-1}xg$ for some $g \in G$; this is an equivalence relation, the **conjugacy relation**, and its classes are the **conjugacy classes** (x is **conjugate** to y).

Exercise 11.8.10

(a) The conjugacy class of 1 is $\{1\}$.
(b) If G is commutative, conjugacy is equality.
(c) How many conjugacy classes does S_3 have?

Solution. (a) $g^{-1}1g = 1$.

(b) $g^{-1}xg = x$.

(c) Three: $\{id\}$, the three transpositions, the two 3-cycles (conjugation preserves cycle type, and $\sigma^{-1}\tau\sigma = \tau^2$, $\tau^{-1}\sigma\tau = \sigma\tau^2$, ...). □

Exercise 11.8.11

(a) For $H \leq G$, $g \in G$: $g^{-1}Hg = \{g^{-1}hg : h \in H\} \leq G$.
(b) On the set $\mathcal{S}$ of subgroups of G, $H \sim K$ iff $K = g^{-1}Hg$ for some g is an equivalence relation (K is a **conjugate subgroup** of H).

Solution. (a) $g^{-1}Hg = \operatorname{Int}(g^{-1})(H)$, the image of a subgroup under a homomorphism.

(b) Reflexive with $g = 1$; symmetric since $H = gKg^{-1}$; transitive since $h^{-1}(g^{-1}Hg)h = (gh)^{-1}H(gh)$. □

Exercise 11.8.12

(a) Give the Cartesian product $G = G_1 \times \cdots \times G_n$ of groups a natural group structure. (b) The i-th **projection** $\pi_i : G \to G_i$ is an epimorphism and the j-th **embedding** $\iota_j : G_j \to G$ is a monomorphism.

Solution. (a) Componentwise multiplication $(g_i)(h_i) = (g_ih_i)$; identity $(1, \ldots, 1)$, inverse (g_i^{-1}).

(b) $\pi_i(g) = g_i$ is a surjective homomorphism; $\iota_j(g) = (1, \ldots, g, \ldots, 1)$ (g in position j) is an injective homomorphism. □

Exercise 11.8.13

(a) $G \times H \approx H \times G$.
(b) $(G \times H) \times K \approx G \times (H \times K) \approx G \times H \times K$.

Solution. $(g, h) \mapsto (h, g)$ and $((g, h), k) \mapsto (g, h, k)$ and $(g, (h, k)) \mapsto (g, h, k)$ are bijective homomorphisms. □

Exercise 11.8.14

$G_1 \times \cdots \times G_n$ is commutative iff all G_i are commutative.

Solution. If all G_i commute, products commute componentwise. Conversely each $G_i \approx \iota_i(G_i)$ is a subgroup of a commutative group. □

Exercise 11.8.15

(a) $\mu_3 \times \mu_3 \not\approx \mu_9$.
(b) $\mu_2 \times \mu_3 \approx \mu_6$.

Solution. (a) Every element of $\mu_3 \times \mu_3$ has order dividing 3, while μ_9 has an element of order 9; orders of elements are preserved by isomorphisms.

(b) $(-1, \omega)$ ($\omega = e^{2\pi i/3}$) has order 6: its powers are $(-1)^k$ and ω^k, which are both trivial only for $2 \mid k$ and $3 \mid k$. So $\mu_2 \times \mu_3$ is cyclic of order 6, hence $\approx \mu_6$ (Exercise 11.7.9). □

Exercise 11.8.16

In Exercise 11.8.12(b): (a) find $\ker \pi_i$ and $\operatorname{im} \iota_j$;
(b) $\pi_i \circ \iota_i = id$;
(c) $\pi_i \circ \iota_j = \mathbf{1}$ for $i \neq j$.

Solution. (a) $\ker \pi_i = \{g : g_i = 1\}$ and $\operatorname{im} \iota_j = \{g : g_k = 1 \text{ for } k \neq j\}$. (b), (c) are direct. □

Remark 11.8.17

For homomorphisms $\varphi, \psi : G \to H$, the pointwise product $(\varphi\psi)(g) = \varphi(g)\psi(g)$ need not be a homomorphism: $(\varphi\psi)(xy) = \varphi(x)\varphi(y)\psi(x)\psi(y)$ differs from $\varphi(x)\psi(x)\varphi(y)\psi(y)$ unless H is commutative. So homomorphisms do not form a group the way linear maps form a vector space.

11.9 Linear groups

Motivation. Groups of invertible matrices and operators are where groups and linear algebra meet. A basis identifies $\mathrm{GL}(V)$ with $\mathrm{GL}_n(F)$ and $\mathrm{SL}(V)$ with $\mathrm{SL}_n(F)$, and an orthonormal basis identifies $\mathbf{O}(V)$ with $\mathbf{O}(n)$. By Cayley's theorem, every finite group is a matrix group.

Example 11.9.1

(a) $\mathrm{GL}_n(F) = \{A \in \mathfrak{M}_{n,n}(F) : A \text{ invertible}\}$, the **general linear group**, is a group under matrix multiplication; $F^\times = \mathrm{GL}_1(F) = F - \{0\}$ is the **unit** (multiplicative) **group** of F.
(b) Subgroups of $\mathrm{GL}_n(F)$ are **linear** (**matrix**) **groups**.
(c) The **special linear group** $\mathrm{SL}_n(F) = \{A \in \mathrm{GL}_n(F) : \det A = 1\} = \ker(\det : \mathrm{GL}_n(F) \to F^\times)$ is a subgroup.

Example 11.9.2

For a finite-dimensional F-vector space V: (a) $\mathrm{GL}(V) = \{L \in \mathfrak{L}(V, V) : L \text{ invertible}\}$ is a group under composition; (b) its subgroups are **linear groups**;
(c) $\mathrm{SL}(V) = \{L \in \mathrm{GL}(V) : \det L = 1\} \leq \mathrm{GL}(V)$.

Example 11.9.3

(a) For $\dim V = n$ and a basis $\mathfrak{B}$, the vector space isomorphism $\Psi_{\mathfrak{B}}^{\mathfrak{B}} : \mathfrak{L}(V,V) \to \mathfrak{M}_{n,n}(F)$, $L \mapsto [L]_{\mathfrak{B}}^{\mathfrak{B}}$, restricts to a group isomorphism $\mathrm{GL}(V) \to \mathrm{GL}_n(F)$: it preserves products ($[LM] = [L][M]$) and maps invertible maps exactly onto invertible matrices.

$$\begin{array}{ccc} \mathrm{GL}(V) & \hookrightarrow & \mathfrak{L}(V,V) \\ {\scriptstyle \Psi_{\mathfrak{B}}^{\mathfrak{B}}}\downarrow & & \downarrow{\scriptstyle \Psi_{\mathfrak{B}}^{\mathfrak{B}}} \\ \mathrm{GL}_n(F) & \hookrightarrow & \mathfrak{M}_{n,n}(F) \end{array}$$

(c) Since $\det L = \det [L]_{\mathfrak{B}}^{\mathfrak{B}}$, it also restricts to $\mathrm{SL}(V) \approx \mathrm{SL}_n(F)$.

Example 11.9.4

If $\dim V = \dim W = n$, then $\mathrm{GL}(V) \approx \mathrm{GL}(W)$. (a) Via bases: $(\Psi_{\mathfrak{C}}^{\mathfrak{C}})^{-1} \circ \Psi_{\mathfrak{B}}^{\mathfrak{B}}$. (b) Via a vector space isomorphism $\varphi : V \to W$ (e.g. with $\varphi(\mathfrak{B}) = \mathfrak{C}$): $\Gamma(L) = \varphi \circ L \circ \varphi^{-1}$. Γ is well defined, $\Gamma(LL') = \varphi L \varphi^{-1} \varphi L' \varphi^{-1} = \Gamma(L)\Gamma(L')$, and $\Gamma^{-1}(M) = \varphi^{-1} M \varphi$. (c) If $\varphi(\mathfrak{B}) = \mathfrak{C}$, the two isomorphisms coincide: $[\varphi L \varphi^{-1}]_{\mathfrak{C}}^{\mathfrak{C}} = [\varphi]_{\mathfrak{B}}^{\mathfrak{C}} [L]_{\mathfrak{B}}^{\mathfrak{B}} [\varphi^{-1}]_{\mathfrak{C}}^{\mathfrak{B}} = [L]_{\mathfrak{B}}^{\mathfrak{B}}$, since $[\varphi]_{\mathfrak{B}}^{\mathfrak{C}} = I$. ("There is only one natural thing.")

Example 11.9.5

$\mu_2 < \mathbb{R}^\times < \mathbb{C}^\times$ and $\mu_n < \mathbb{C}^\times$.

Example 11.9.6

Theorem 1.3.6(3) (every invertible matrix is a product of elementary matrices) says: $\mathrm{GL}_n(F)$ is generated by elementary matrices.

Example 11.9.7

$\mathbf{T}_n(F) = \{A \in \mathrm{GL}_n(F) : A \text{ upper triangular}\}$ and $\mathbf{V}_n(F) = \{A \in \mathbf{T}_n(F) : A \text{ unipotent}\}$ (diagonal entries 1) are matrix groups.

Proof [Supplement]. Products of upper triangular matrices are upper triangular with diagonal entries multiplied, and the inverse of an invertible upper triangular matrix is upper triangular (back substitution) with inverted diagonal entries. So both sets are closed under products and inverses and contain I. ■

Exercise 11.9.8

$(F,+)$ is a matrix group: $(F,+) \approx \mathbf{V}_2(F)$.

Solution. $a \mapsto \left(\begin{smallmatrix} 1 & a \\ 0 & 1 \end{smallmatrix}\right)$ is a bijection, and $\left(\begin{smallmatrix} 1 & a \\ 0 & 1 \end{smallmatrix}\right)\left(\begin{smallmatrix} 1 & b \\ 0 & 1 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 1 & a+b \\ 0 & 1 \end{smallmatrix}\right)$. □

Exercise 11.9.9

The group $(F^3, \bullet)$ of Exercise 11.1.14 is isomorphic to $\mathbf{V}_3(F)$.

Solution. $(a_1,a_2,a_3)^t \mapsto \left(\begin{smallmatrix} 1 & a_1 & a_2 \\ 0 & 1 & a_3 \\ 0 & 0 & 1 \end{smallmatrix}\right)$; the product of two such matrices has entries $a_1 + b_1$, $a_2 + b_2 + a_1 b_3$, $a_3 + b_3$, which is the formula for $\bullet$. (So associativity there came for free.) □

Example 11.9.10

Conjugacy in $\mathrm{GL}_n(F)$ is the similarity relation of §5.5.

Exercise 11.9.11

If $L \in \mathrm{GL}(V)$ is unipotent and $L \neq I$, then $|L| = \infty$ in $\mathrm{GL}(V)$ (over $F = \mathbb{R}$ or $\mathbb{C}$).

Solution. Write $L = I + N$ with N nilpotent, $N \neq 0$, and let $k \geq 1$ be largest with $N^k \neq 0$. Then $L^m = \sum_{j \leq k} \binom{m}{j} N^j$, and the coefficient of N in $L^m - I = mN + \dots$ is $m \neq 0$. Since $I, N, \dots, N^k$ are linearly independent (if $\sum c_j N^j = 0$ with c_r the first nonzero coefficient, multiplying by N^{k-r} gives $c_r N^k = 0$), $L^m \neq I$ for every $m \geq 1$. □

Exercise 11.9.12

For fixed $C \in \mathrm{GL}_n(\mathbb{R})$, $\varphi(A) = CAC^{-1}$ and $\psi(A) = (A^t)^{-1}$ belong to $\mathrm{Aut}(\mathrm{SL}_n(\mathbb{R}))$. If $C \notin \mathrm{SL}_n(\mathbb{R})$, are φ, ψ outer automorphisms?

Solution. Both preserve $\mathrm{SL}_n(\mathbb{R})$ (det is unchanged) and products ($((AB)^t)^{-1} = (A^t)^{-1}(B^t)^{-1}$), and are bijective ($\psi^2 = id$, $\varphi^{-1} = \mathrm{Int}(C^{-1})$).

[Supplement] φ. An element $C' \in \mathrm{SL}_n(\mathbb{R})$ induces the same automorphism iff $C^{-1}C'$ commutes with all of $\mathrm{SL}_n(\mathbb{R})$, i.e. is a scalar λI (it must commute with all elementary matrices of type 3). So φ is inner iff $\lambda^n \det C = 1$ for some real λ. If n is odd this is solvable, and φ is inner. If n is even and $\det C < 0$, it is not, and φ is outer.

ψ. For $n = 2$, ψ is inner: $(A^t)^{-1} = JAJ^{-1}$ with $J = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{R})$. For $n \geq 3$, ψ is outer, because inner automorphisms preserve the trace. $A = \mathrm{diag}(2, 3, \frac{1}{6}, 1, \dots, 1)$ has trace $2 + 3 + \frac{1}{6} + (n-3)$, while $\psi(A) = \mathrm{diag}(\frac{1}{2}, \frac{1}{3}, 6, 1, \dots, 1)$ has trace $\frac{1}{2} + \frac{1}{3} + 6 + (n-3)$. □

Exercise 11.9.13

For the invertible diagonal matrices $\mathbf{D}_n(F)$: (a) $\mathbf{D}_n(F) \approx F^\times \times \cdots \times F^\times$ (n factors);
(b) $\mathbf{D}_n(F) \cap \mathrm{SL}_n(F) \approx (F^\times)^{n-1}$ for $n \geq 2$.

Solution. (a) $\mathrm{diag}(a_1, \dots, a_n) \mapsto (a_1, \dots, a_n)$.

(b) The last entry is determined by $a_n = (a_1 \cdots a_{n-1})^{-1}$; project to the first $n-1$ entries. □

Exercise 11.9.14

(a) If $m \leq n$, $\mathrm{GL}_m(F)$ is isomorphic to a subgroup of $\mathrm{GL}_n(F)$.
(b) If $n_1 + \cdots + n_r = n$, $\mathrm{GL}_{n_1}(F) \times \cdots \times \mathrm{GL}_{n_r}(F)$ is isomorphic to a subgroup of $\mathrm{GL}_n(F)$.

Solution. (b) $(A_1, \dots, A_r) \mapsto A_1 \oplus \cdots \oplus A_r$ (block diagonal) is a monomorphism. (a) is the case $A \mapsto A \oplus I_{n-m}$. □

Exercise 11.9.15

(a) $\mathbf{SO}(n) < \mathbf{O}(n) < \mathrm{GL}_n(\mathbb{R})$;
(b) $\mathbf{SU}(n) < \mathbf{U}(n) < \mathrm{GL}_n(\mathbb{C})$;
(c) $\mathbf{SO}(n) = \mathrm{SL}_n(\mathbb{R}) \cap \mathbf{O}(n)$;
(d) $\mathbf{SU}(n) = \mathrm{SL}_n(\mathbb{C}) \cap \mathbf{U}(n)$.

Solution. By Exercises 9.3.3, 9.3.10, 10.6.10 and the definitions. □

Example 11.9.16

With $\mathbf{O}(\mathbb{R}^n) = \mathbf{O}(\mathbb{R}^n, \text{dot product})$, $\Psi_{\mathcal{E}}^{\mathcal{E}}$ restricts to $\mathbf{O}(\mathbb{R}^n) \approx \mathbf{O}(n)$ (by the definition of $\mathbf{O}(n)$). For any orthonormal basis $\mathfrak{B}$ of $\mathbb{R}^n$, $\Psi_{\mathfrak{B}}^{\mathfrak{B}}$ also restricts to $\mathbf{O}(\mathbb{R}^n) \approx \mathbf{O}(n)$, by Remark 10.6.4: $\mathbf{O}(n) = \{[L]_{\mathfrak{B}}^{\mathfrak{B}} : L \in \mathbf{O}(\mathbb{R}^n)\}$ (see Exercise 9.6.11).

Exercise 11.9.17

(a) $\mathbf{U}(\mathbb{C}^n) \approx \mathbf{U}(n)$.
(b) $\mathbf{SO}(\mathbb{R}^n) \approx \mathbf{SO}(n)$.
(c) Define $\mathbf{SU}(\mathbb{C}^n)$ and show $\mathbf{SU}(\mathbb{C}^n) \approx \mathbf{SU}(n)$.

Solution. As in Example 11.9.16, restricting $\Psi_{\mathcal{E}}^{\mathcal{E}}$; for (b), (c) also $\det L = \det[L]_{\mathcal{E}}^{\mathcal{E}}$, with $\mathbf{SU}(\mathbb{C}^n) = \{L \in \mathbf{U}(\mathbb{C}^n) : \det L = 1\}$. □

Example 11.9.18

If $\varphi : V \to W$ is an inner product space isomorphism, $\Gamma(L) = \varphi L \varphi^{-1}$ (Example 11.9.4) restricts to $\mathbf{O}(V) \approx \mathbf{O}(W)$ (resp. $\mathbf{U}$), since L preserves inner products iff $\varphi L \varphi^{-1}$ does (orthogonal operators are the inner product space automorphisms).

Example 11.9.19

The **circle group** $S^1 = \{X \in \mathbb{R}^2 : \|X\| = 1\} = \{z \in \mathbb{C} : |z| = 1\} \leq \mathbb{C}^\times$, and $\varphi : \mathbf{SO}(2) \to S^1$, $\varphi(R_\theta) = e^{i\theta}$, is a group isomorphism.

Exercise 11.9.20

$\mathbf{SO}(2) \approx S^1 \approx \mathbf{U}(1)$.

Solution. $\varphi(R_\theta R_\eta) = \varphi(R_{\theta+\eta}) = e^{i(\theta+\eta)} = e^{i\theta}e^{i\eta}$, and φ is bijective (Theorem 9.5.2). $\mathbf{U}(1) = \{z \in \mathbb{C} : \bar{z}z = 1\} = S^1$. □

Exercise 11.9.21

$S_n \approx \{I_\sigma \in \mathfrak{M}_{n,n}(F) : \sigma \in S_n\} \approx \{P_\sigma \in \mathfrak{L}(V,V) : \sigma \in S_n\}$ ($\dim V = n$), via permutation matrices and permutation operators.

Solution. The permutation matrix I_σ (with $I_\sigma e_j = e_{\sigma(j)}$) satisfies $I_\sigma I_\tau = I_{\sigma\tau}$, and $\sigma \mapsto I_\sigma$ is injective; similarly for P_σ on a basis. □

Corollary 11.9.22

Every finite group is isomorphic to a linear (matrix) group.

Proof. By Cayley's theorem G embeds in $S_G \approx S_n$ ($n = |G|$), and S_n embeds in $\mathrm{GL}_n(F)$ by Exercise 11.9.21. ■

(Representing a group by matrices is a *linear representation*; by permutations, a *permutation representation*.)

Example 11.9.23

In $\mathbf{O}(2)$ let $R = R_{2\pi/3}$, $S = \left(\begin{smallmatrix} -1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$, and $D_3 = \langle R, S\rangle \leq \mathbf{O}(2)$.
(a) By the plane rule $S^{-1}RS = R^{-1} = R^2$, i.e. $RS = SR^2$. Since $\{R, S\} \cup \{R, S\}^{-1} = \{R, R^2, S\}$, D_3 consists of words in R, R^2, S; moving every S to the left with $RS = SR^2$ gives $D_3 = \{S^i R^j : 0 \leq i \leq 1,\ 0 \leq j \leq 2\}$, and $|D_3| = 6$ (the six matrices are distinct: compare determinants and first columns). The relations $S^2 = I$, $R^3 = I$, $RS = SR^2$ determine the table; e.g. $(SR)(SR^2) = S(RS)R^2 = S^2R^4 = R$.
(b) These are exactly the relations of $S_3 = \langle \sigma, \tau\rangle$ with $\sigma^2 = 1$, $\tau^3 = 1$, $\tau\sigma = \sigma\tau^2$ (Example 11.8.4).
(c) So $\varphi(\sigma^i\tau^j) = S^iR^j$ is a bijection $S_3 \to D_3$ preserving the multiplication tables (which both relations completely determine): $S_3 \approx D_3$.

Exercise 11.9.24

For $n \geq 3$, the **dihedral group** $D_n = \langle R_{2\pi/n}, S\rangle \leq \mathbf{O}(2)$ has $|D_n| = 2n$.

Solution. With $R = R_{2\pi/n}$: $R^n = I$, $S^2 = I$, $RS = SR^{-1}$, so as in Example 11.9.23 $D_n = \{S^iR^j : 0 \leq i \leq 1, 0 \leq j < n\}$. These $2n$ elements are distinct: $\det(S^iR^j) = (-1)^i$ separates i, and R^j, $0 \leq j < n$, are distinct rotations. □

Exercise 11.9.25

(a) For $\sigma = (2\,5)(3\,4)$, $\tau = (1\,2\,3\,4\,5) \in S_5$, $G_5 = \langle \sigma, \tau\rangle \approx D_5$. (b) For $\sigma = (2\,6)(3\,5)$, $\tau = (1\,2\,3\,4\,5\,6) \in S_6$, $G_6 = \langle \sigma, \tau\rangle \approx D_6$. (c) Interpret (a),
(b) as permutations of the vertices of a regular pentagon/hexagon, and generalize to the regular n-gon.

Solution. In both cases $\sigma^2 = 1$, $\tau^n = 1$, and $\sigma\tau\sigma = \tau^{-1}$ (check on 1: $\sigma\tau\sigma(1) = \sigma\tau(1) = \sigma(2) = n = \tau^{-1}(1)$, and similarly on the other points). So $\sigma^i\tau^j \mapsto S^iR^j$ is an isomorphism onto D_n, as in Example 11.9.23; both groups have $2n$ elements. Geometrically, numbering the vertices of a regular n-gon $1, \ldots, n$ counterclockwise, τ is the rotation by $2\pi/n$ and σ the reflection in the axis through vertex 1; D_n is the symmetry group of the regular n-gon. □